Computer Architecture- Quiz #5-G

Name: ______________________________ **Roll#:**______________________

Section:_______

Total Time: 20 minutes Total Marks: 20 7th April, 2026

CLO 2: **Question 1** **Marks 10**

Consider the algorithm given below run it for value where A =10100, B = 11

A/B = Q + R/B	$R' = 0$ for i = N-1 to 0 $\quad R = \{R' << 1, A_i\}$ $\quad D = R - B$ $\quad$ if $D < 0$, $Q_i = 0$; $R' = R$ $\quad$ else $Q_i = 1$; $R' = D$ $R = R'$

Initialization	**A =** , **B =** , **Q_i =** **R' =** , **N =** , **R =** , **D =**
N = 4	**A =** , **B =** , **Q_i =** **R' =** , **N =** , **R =** , **D =**
N = 3	**A =** , **B =** , **Q_i =** **R' =** , **N =** , **R =** , **D =**
N = 2	**A =** , **B =** , **Q_i =** **R' =** , **N =** , **R =** , **D =**
N = 1	**A =** , **B =** , **Q_i =** **R' =** , **N =** , **R =** , **D =**
N = 0	**A =** , **B =** , **Q_i =** **R' =** , **N =** , **R =** , **D =**

Final $Q_{4:0}$= , **R=**

CLO 2: **Question 2** **Marks 5**

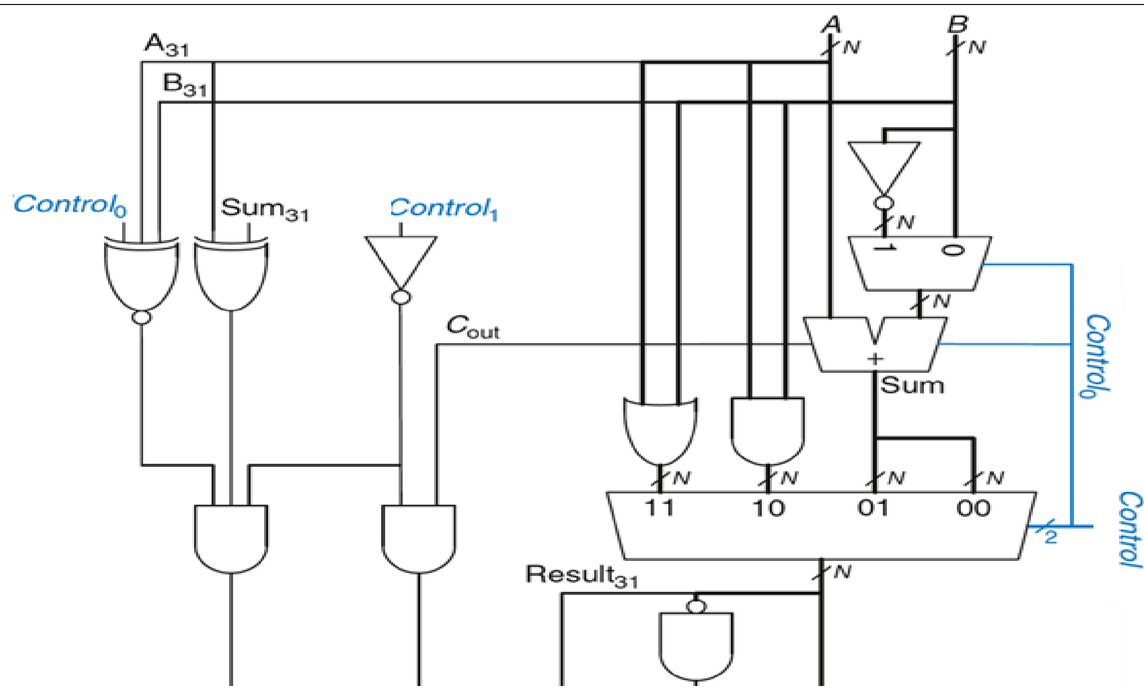
Consider **Carry-Lookahead Adder** and Write generic single Bit expressions for generate (G_i), propagate (P_i) signals and Carryout signals.

G_i = , P_i =

C_i =

CLO 2: **Question 3** **Marks 5**

Identify the **type of circuit** given below. Name all the unidentified inputs and outputs. Identify **CONTROL** bits and based on control bits fill the given **table**



CONTROL 1:0	FUNCTION

Circuit Name: